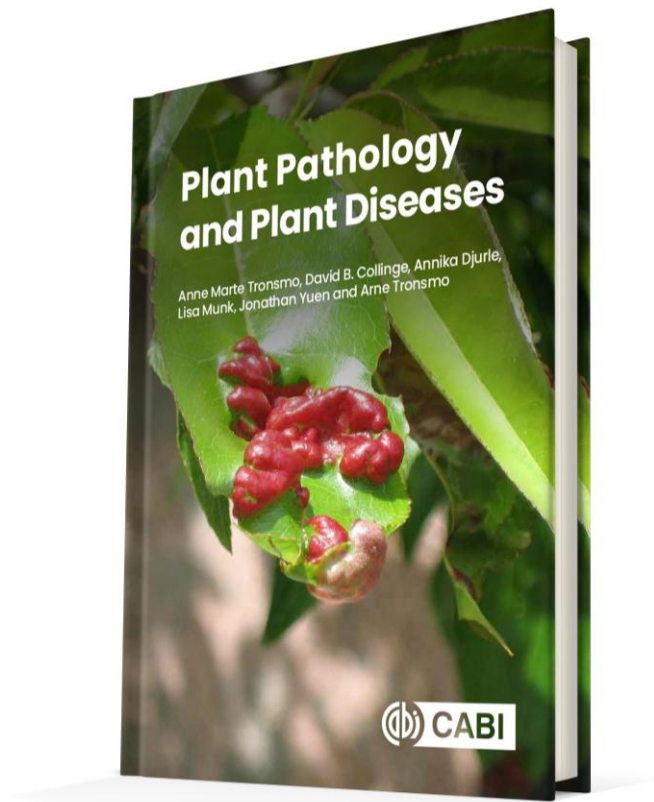




Plant Pathology and Plant Diseases

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Lisa Munk, Jonathan Yuen and Arne Tronsmo 2020

Fig. 7.1 Examples of virus-induced symptoms. (A) Tobacco rattle virus. (B) Plum pox virus in plum. (C) Necrosis and mosaic in begonia flowers caused by Impatiens necrotic spot virus. (© D-R. Blystad.)

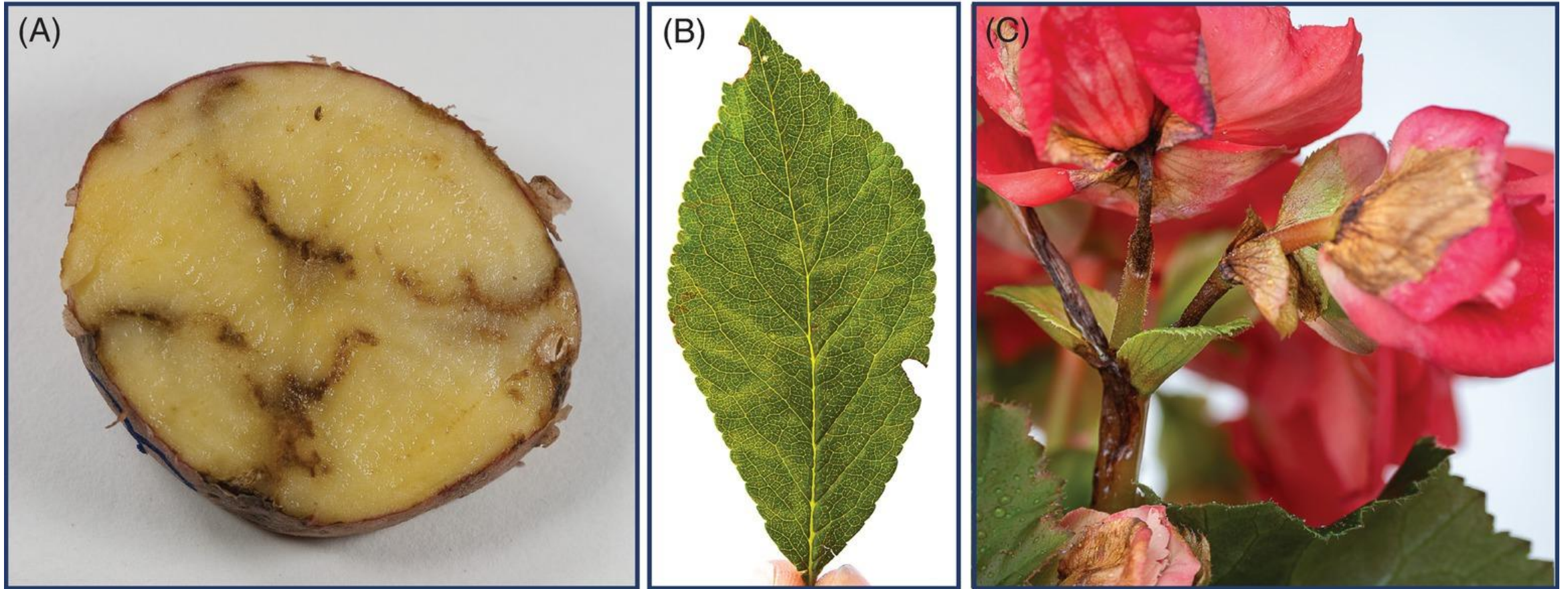


Fig. 7.2 Cucumber mosaic virus causing yellow mosaic in *Echinacea purpurea*. (© D-R. Blystad.)



Fig. 7.3 Apple mosaic virus giving typical symptoms in apple leaves. (© D-R. Blystad.)



Fig. 7.4 Yellow leaves on barley caused by the luteovirus barley yellow dwarf virus. (© A. Kvarnheden.)



Fig. 7.5 A nepovirus – cherry leafroll virus – is associated with mosaic in birch. (© J. Valkonen.)



Fig. 7.6 Brown arches and spots (spraing) in potato tubers caused by potato mop-top virus.

(T. Munthe © NIBIO.)



Fig. 7.7 Pepino mosaic virus giving yellow spots and mosaic in tomato. (© D-R. Blystad.)



**Fig. 7.8 Potato virus Y in potato
giving necrosis on the veins. (© D-R.
Blystad.)**



Fig. 7.9 Tobacco mosaic virus in tobacco – *Nicotiana tabacum*. (© D-R. Blystad.)



Fig. 7.10 Tobacco rattle virus causing mosaic in the ornamental perennial *Epimedium* sp. (© D-R. Blystad.)



Fig. 7.11 Strawberry crinkle virus giving clear symptoms in a sensitive indicator of *Fragaria*.
(© D-R. Blystad.)



Fig. 7.12 Yellow stripes and necrosis caused by wheat dwarf virus in wheat. (© A. Kvarnheden.)

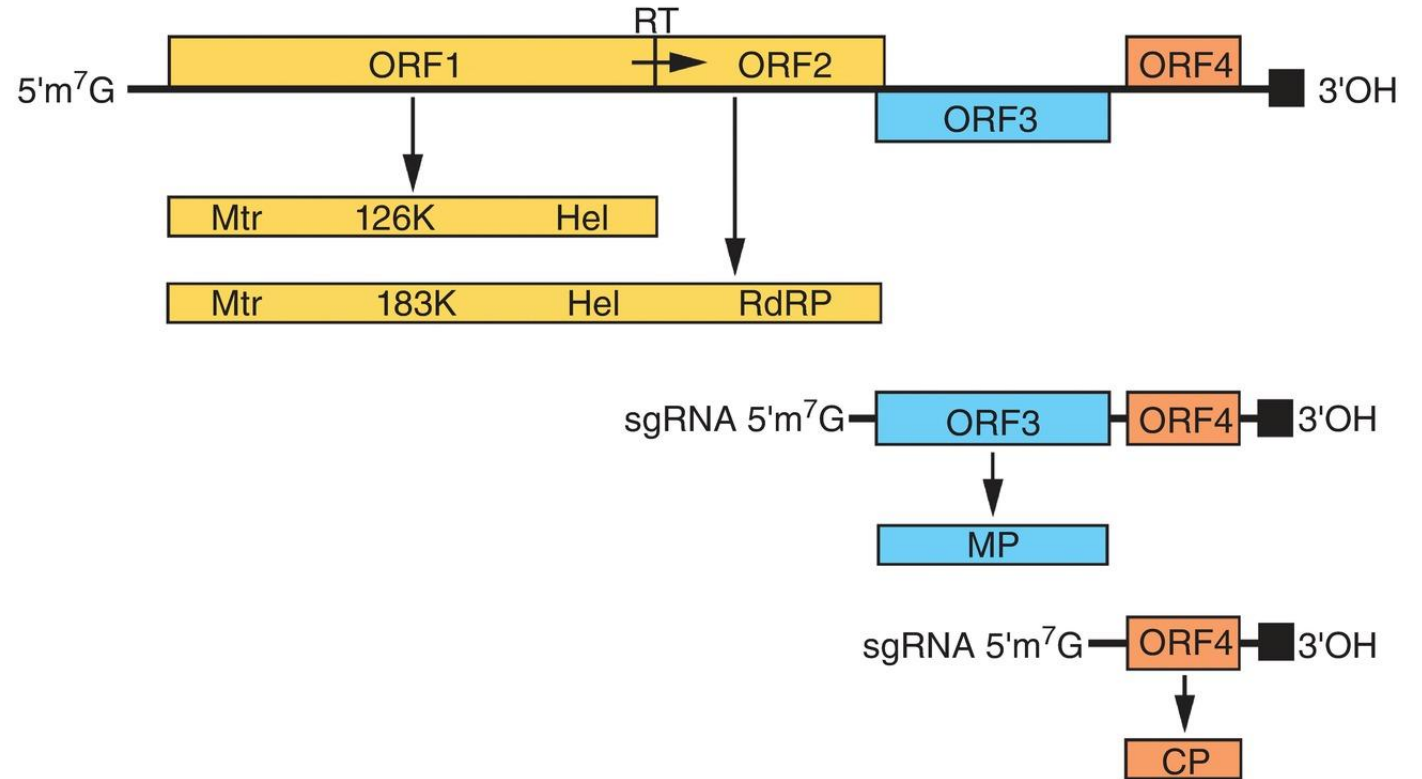


Fig. 7.13 Tomato spotted wilt virus in tomato – necrotic bronze coloured ringspots and bands. (© D-R. Blystad.)



Fig. 7.14 Strategies for plant virus translation. (A) The strategy with sub-genomic RNAs is used by tobacco mosaic virus. (B) Polyprotein strategy: the genome of potato virus Y. (A: Adams et al., 2017; B Wylie et al., 2019.)

(A) Tobacco mosaic virus, TMV (6,395 nts)



(B) Potato virus Y, PVY (9703 nts)

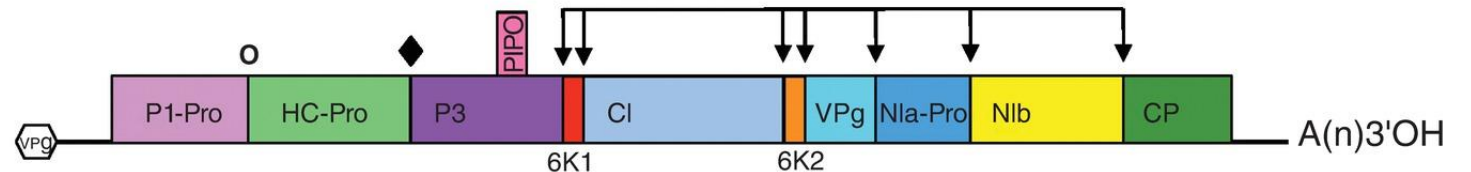


Fig. 7.15 Mosaic caused by viruses in some instances been regarded as giving ornamental value. (A) Mosaic in *Abutilon* caused by abutilon mosaic virus. (B) Tulip with petal stripes. (© D-R. Blystad.)

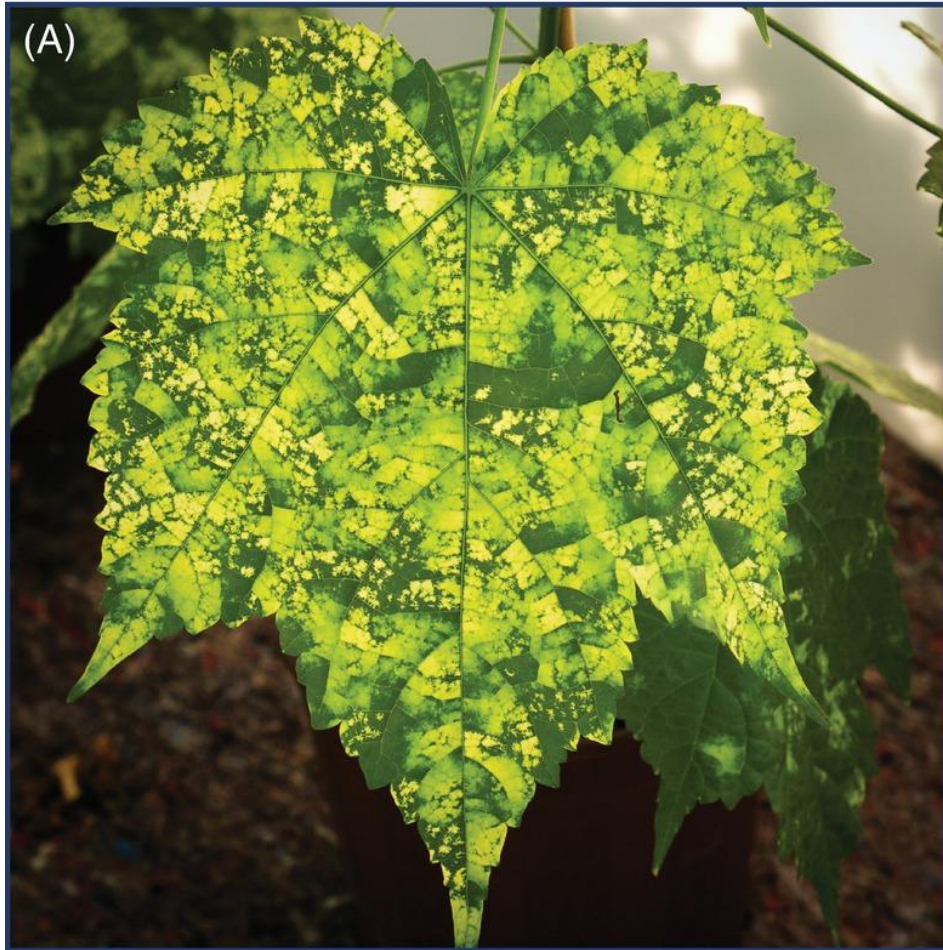


Fig. 7.16 Diagnosis and detection of plant viruses are dependent on different aspects related to the virus. (© D-R. Blystad.)

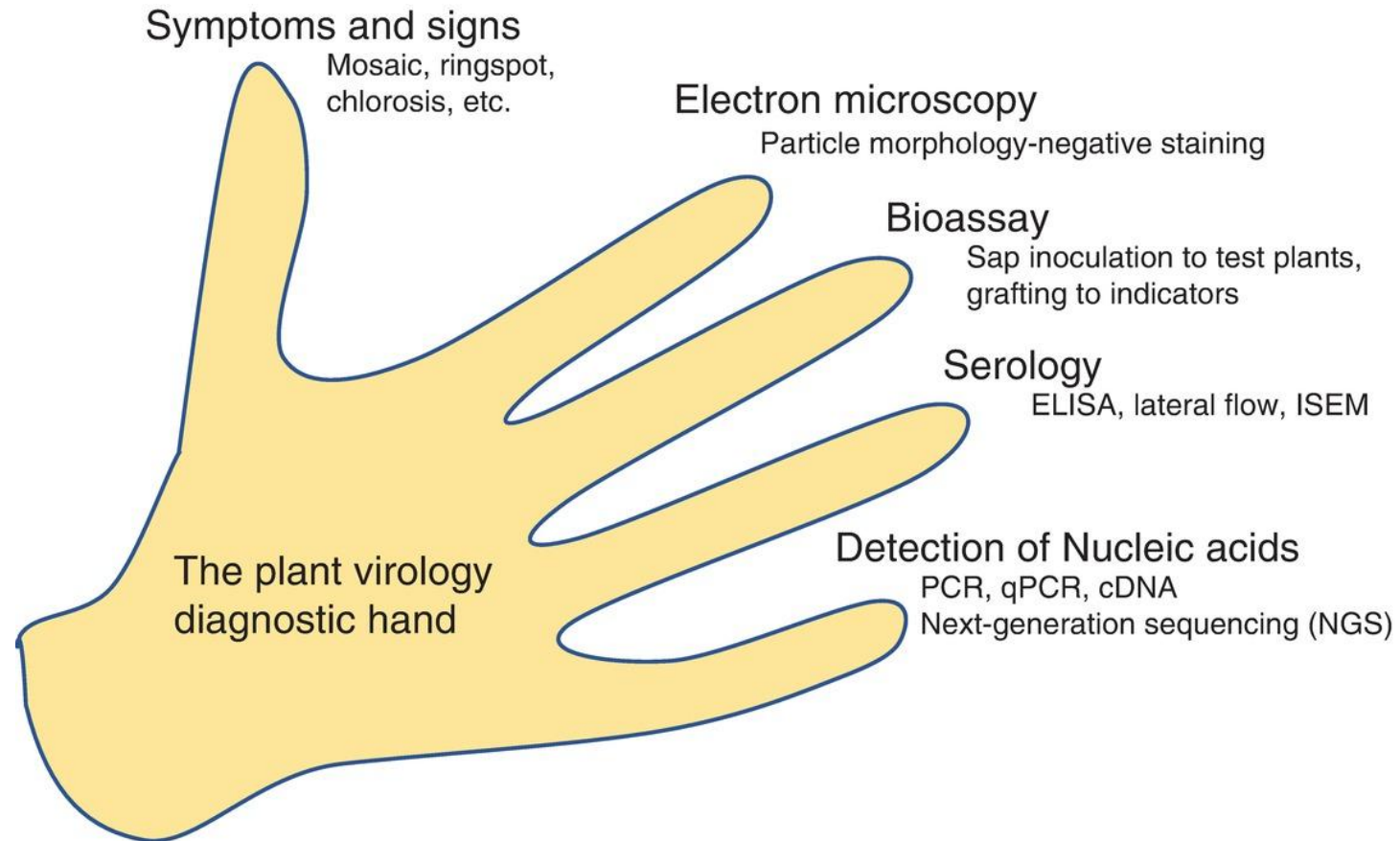


Fig. 7.17 Symptoms caused by viruses on test plants mechanically inoculated with virus. (A) Lettuce mosaic virus giving local lesions in *Chenopodium quinoa*. (B) Lettuce mosaic virus giving systemic mosaic in *C. quinoa* some days later. (C) Tobacco rattle virus giving local lesions in *Nicotiana tabacum*. (© D-R. Blystad.)



Fig. 7.18 Virus particles. (A) Pelargonium flower break virus (B) Potato virus X. (C) Lettuce mosaic virus (D) Cucumber green mottle mosaic virus. (A: © S. Haugslie; B-C: © D-R Blystad; D: © L.C. Hermansen)

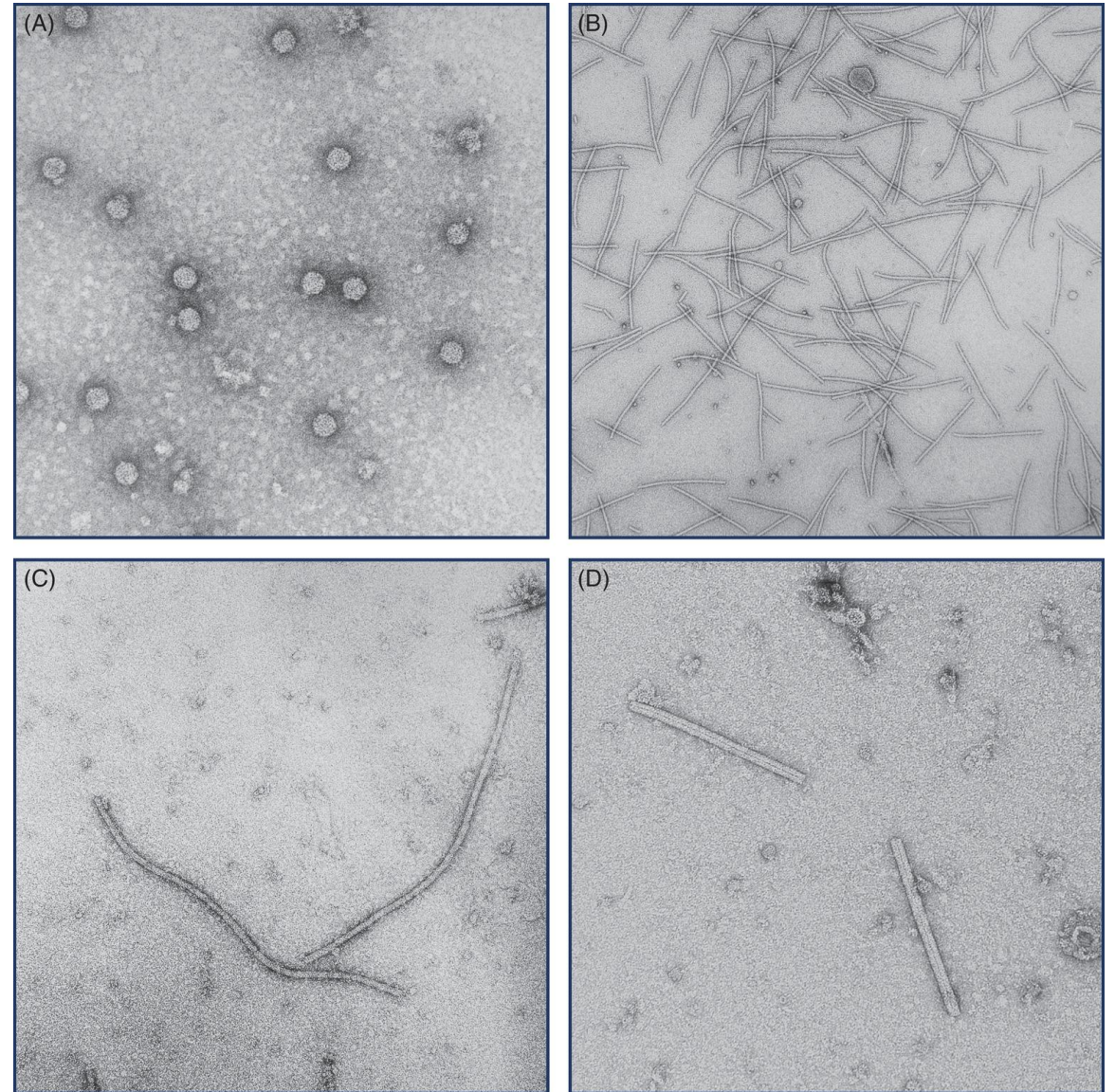


Fig. 7.19 Important aphid vectors. (A) The green peach aphid (*Myzus persicae*) is the aphid species known to vector most different virus species. (B) Bird cherry-oat aphid (*Rhopalosiphum padi*) occurs in high numbers in springtime and is an important virus vector. (A: E. Fløistad © NIBIO. B: D-R Blystad.)



Fig. 7.20 Plant virus epidemiology often involves vectors. This makes an additional and complicating factor in the understanding of plant virus epidemics. (© D-R. Blystad.)

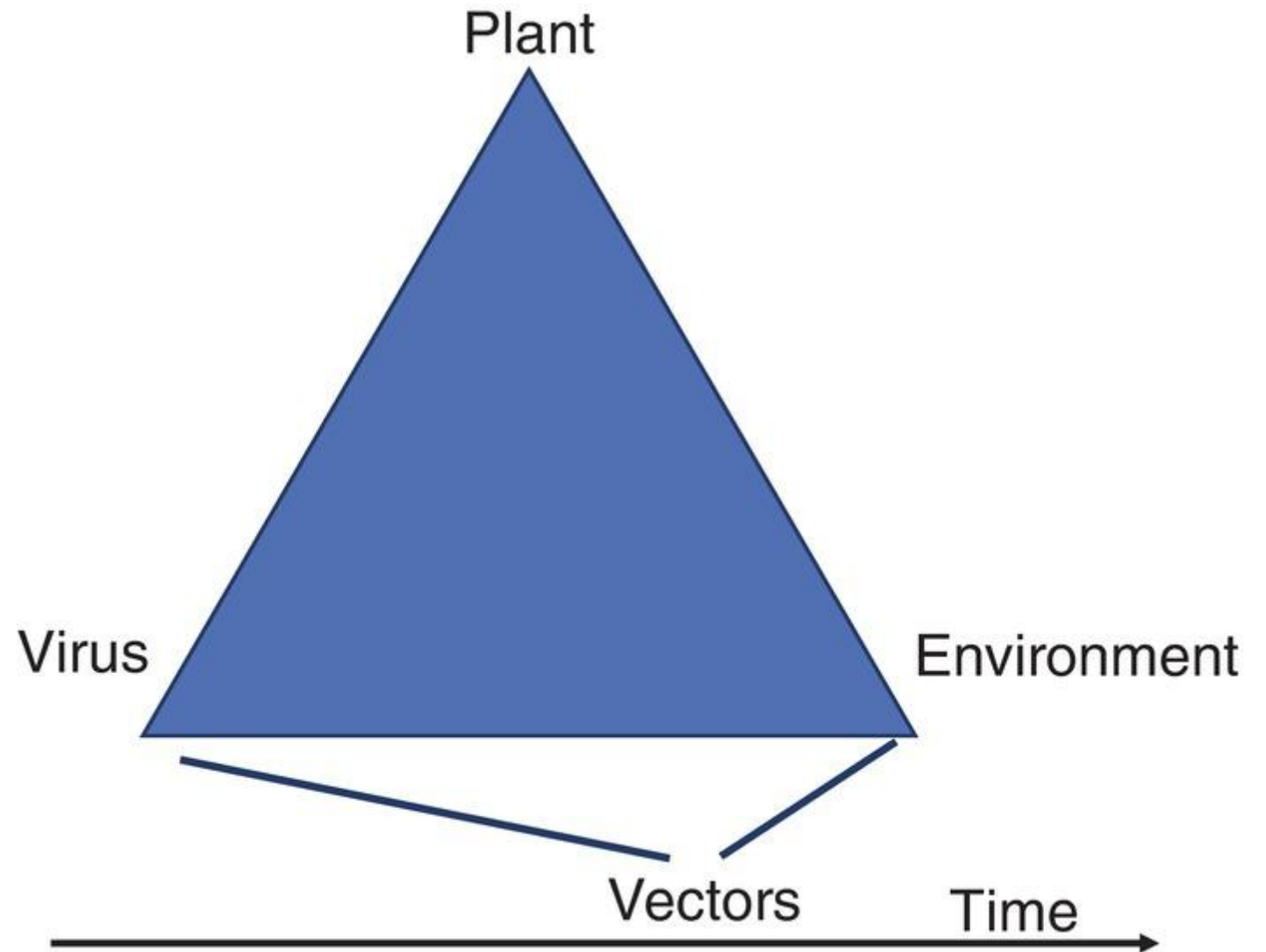


Fig. 7.21 A nuclear stock programme has to establish a few virus-free nuclear stock plants as a foundation for the further propagation stocks. . (© D-R. Blystad.)

Nuclear stock programmes - a general outline

